

Math 514, F13
Quiz 3

Name: Answer
SSN: xxx-xx-

Instructions: You must show all of your work to receive a credit for a correct answer.

Problem 1. (4 points) For a standard normal random variable Z , find $P(|Z| < 1.66)$.

$$\begin{aligned}P(|Z| < 1.66) &= P(-1.66 < Z < 1.66) \\&= P(Z < 1.66) - P(Z \leq -1.66) \\&= \Phi(1.66) - P(Z \geq 1.66) \\&= \Phi(1.66) - (1 - P(Z \leq 1.66)) \\&= 2\Phi(1.66) - 1 \\&\approx 2 \times 0.9515 - 1 = 0.903\end{aligned}$$

Problem 2. (6 points) Suppose that the amount of time that a certain battery functions is a normal random variable with mean 400 hours and standard deviation 40 hours. Suppose that an individual owns two such batteries. What is the probability that the total life of the batteries will exceed 750 hours?

Let X_1 and X_2 denote the life of the first and second batteries.
Then $X_1 + X_2$ is a normal random variable with mean $\mu = \mu_1 + \mu_2 = 800$ hours and standard deviation $\sigma = \sqrt{40^2 + 40^2} = 40\sqrt{2}$. Thus,

$$\begin{aligned}P\{X_1 + X_2 > 750\} &= P\left\{\frac{X_1 + X_2 - 800}{40\sqrt{2}} > \frac{750 - 800}{40\sqrt{2}}\right\} \\&= P\left\{Z > -\frac{5}{4\sqrt{2}}\right\} \\&\approx P\{Z > -0.8839\} \\&= P\{Z < 0.8839\} \\&= \Phi(0.8839) \\&\approx 0.8116\end{aligned}$$